

$$\begin{aligned}
 15) \quad x_1 + 3x_3 &= 2 \\
 x_2 - 3x_4 &= 3 \\
 -2x_2 + 3x_3 + 2x_4 &= 1 \\
 3x_1 + 7x_4 &= -5
 \end{aligned}$$

Augmented matrix:
$$\left[\begin{array}{cccc|c} 1 & 0 & 3 & 0 & 2 \\ 0 & 1 & 0 & -3 & 3 \\ 0 & -2 & 3 & 2 & 1 \\ 3 & 0 & 0 & 7 & -5 \end{array} \right]$$

$$\text{Row 4} = \text{Row 4} - 3 \times \text{Row 1}$$

$$\text{Row 3} = \text{Row 3} + 2 \times \text{Row 2}$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 3 & 0 & 2 \\ 0 & 1 & 0 & -3 & 3 \\ 0 & 0 & 3 & -4 & 7 \\ 0 & 0 & -9 & 7 & -11 \end{array} \right]$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 3 & 0 & 2 \\ 0 & 1 & 0 & -3 & 3 \\ 0 & -2 & 3 & 2 & 1 \\ 0 & 0 & -9 & 7 & -11 \end{array} \right]$$

$$\text{Row 4} = \text{Row 4} + 3 \times \text{Row 3}$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 3 & 0 & 2 \\ 0 & 1 & 0 & -3 & 3 \\ 0 & 0 & 3 & -4 & 7 \\ 0 & 0 & 0 & -5 & 10 \end{array} \right]$$

In equation form we can write

$$\begin{aligned}
 \textcircled{1} \quad x_1 + 3x_3 &= 2 & \textcircled{2} \quad x_2 - 3x_4 &= 3 & \textcircled{3} \quad 3x_3 - 4x_4 &= 7 & \textcircled{4} \quad -5x_4 &= 10
 \end{aligned}$$

From $\textcircled{4}$, we get a unique value of x_4 . From $\textcircled{3}$, substituting x_4 , we get a value of x_3 (unique). From $\textcircled{2}$, substituting x_4 , we get a unique value of x_2 . From $\textcircled{1}$, substituting x_3 , we get a unique value of x_1 .

$$\begin{aligned}
 16) \quad x_1 + 2x_2 + 2x_3 - 2x_4 &= -3 \\
 x_3 + 3x_4 &= 1 \\
 -2x_1 + 3x_2 + 2x_3 + x_4 &= 5
 \end{aligned}$$

Augmented matrix:
$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & -2 & -3 \\ 0 & 2 & 2 & 0 & 0 \\ 0 & 0 & 1 & 3 & 1 \\ -2 & 3 & 2 & 1 & 5 \end{array} \right]$$

$$\text{Row 4} = \text{Row 4} + 2 \times \text{Row 1}$$

$$\text{Row 2} = \text{Row 2} \times \frac{1}{2}$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & -2 & -3 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 3 & 1 \\ 0 & 3 & 2 & -3 & -1 \end{array} \right]$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & -2 & -3 \\ 0 & 2 & 2 & 0 & 0 \\ 0 & 0 & 1 & 3 & 1 \\ 0 & 3 & 2 & -3 & -1 \end{array} \right]$$

$$\text{Row 4} = \text{Row 4} - 3 \times \text{Row 3}$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & -2 & -3 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 3 & 1 \\ 0 & 0 & -1 & -3 & -1 \end{array} \right]$$

$$\text{Row 4} = \text{Row 4} + \text{Row 3}$$